THE TECHNOLOGY

ECLSS SUBSYSTEMS
20+ Monitored Parameters

Atmosphere Revitalization
O■ · CO■ · Humidity

Oxygen Generation System
Output Rate · O■ Purity

Water Recovery System
Purity · Production Rate

Temp & Humidity Control
Cabin Temperature

Trace Contaminant Control
NH■ · H■ · CO

Pressure Control
Cabin Pressure

Microbial Monitoring
Bacterial / Fungal Count

Mass Spectrometer
N■ · O■ · CO■ · CH■

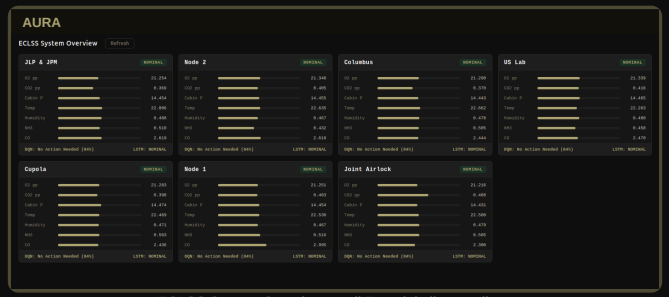
ML PIPELINE
Four-Model Ensemble

Isolation Forest
Unsupervised multi-sensor outlier detection

Random Forest
Supervised fault type classification

LSTM Network
Temporal drift & trend sequence modeling

DQN Agent
RL-based adaptive alert suppression

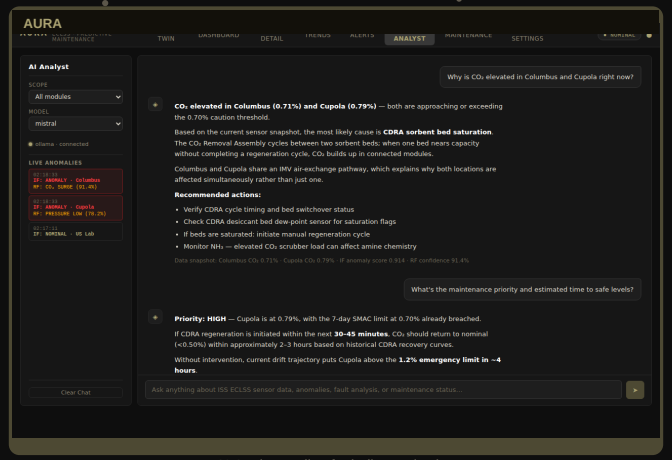


ECLSS System Overview — all 7 module live readings

INTELLIGENCE LAYER

AI ANALYST
Conversational System Intelligence

- Natural language interface to all live ECLSS sensor data
- Powered by Ollama (local) or Groq cloud LLM backend
- Context-aware across all 7 ISS module locations
- Auto-appends live anomaly & sensor snapshot per query



AI Analyst — live fault diagnosis chat

DIGITAL TWIN
3D Interactive ISS Visualization

- Real-time Three.js 3D ISS model with anomaly overlays
- Color-coded module health tied to live sensor data
- Orbit-controls camera with anomaly pulse indicators

Digital Twin — 3D ISS model with live status

AURA

ECLSS PREDICTIVE MAINTENANCE

Mission-Critical Intelligence
for the International Space Station

Real-time AI-driven monitoring, anomaly detection, and predictive maintenance for the ISS ECLSS life support systems.

AI-Powered · Real-Time · 7 Modules · LSTM + DQN · Digital Twin · Open Source

7	4	20+	1s
ISS Modules	ML Models	Sensors	Tick Rate

AURA

ECLSS PREDICTIVE MAINTENANCE

AURA brings space-grade AI to the front line of life support reliability. From real-time anomaly detection to conversational AI fault analysis, AURA gives mission teams the insight they need before the alarm sounds.

SCAN TO VISIT
aurahunch.space

Website	aurahunch.space
Runtime	Python · FastAPI · SQLite
AI Backend	Ollama (local)

Intelligent Life Support.

Open-source · ISS ECLSS simulation

HOW IT WORKS

01 Data Generation
Background engine samples all 7 ISS locations every second with realistic sensor simulation and optional fault injection.

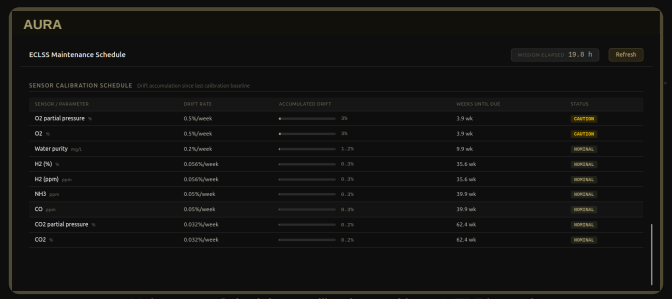
02 ML Inference
Four-model ensemble runs each tick: Isolation Forest detects outliers, Random Forest classifies fault type, LSTM tracks drift, DQN controls alerting.

03 Alert Triggering
After N consecutive anomalous ticks, AURA raises a severity-graded alert with location, parameter, and recommended action.

04 Trend Analysis
Built-in trend detector identifies slopes and deviations across any sensor or module before alert thresholds are reached.

05 AI Query
Natural language question triggers a live sensor snapshot that is appended to an LLM prompt for contextual fault diagnosis.

06 Maintenance Action
Scheduler aggregates MTBF intervals and calibration drift, giving operators a clear timeline of upcoming ECLSS service tasks.



Maintenance Schedule — calibration tracking & MTBF intervals

MONITORED LOCATIONS

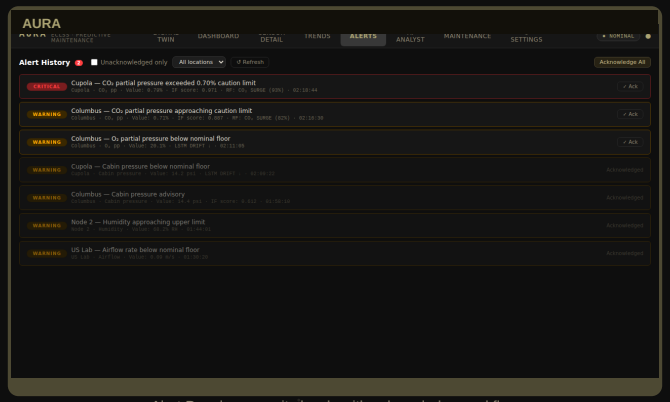
JLP & JPM · Node 2 · Columbus · US Lab
Cupola · Node 1 · Joint Airlock

ECLSS System Overview — all 7 module live readings

KEY FEATURES

REAL-TIME DASHBOARD
Live Sensor Monitoring

- Live WebSocket sensor readings across all locations
- Sensor drill-down with 60/100/200/1000-point windows
- Alert badge counts on nav tabs with auto-refresh



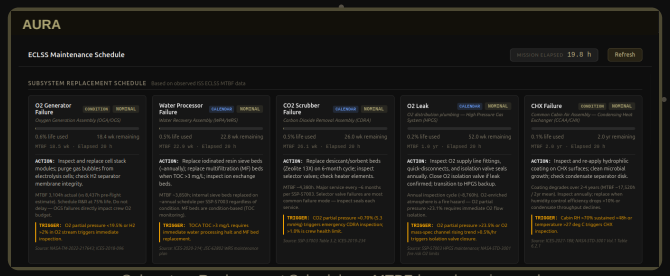
Alert Panel — severity levels with acknowledge workflow

ALERT SYSTEM
Configurable & Actionable

- NOMINAL · ADVISORY · WARNING · CRITICAL severity grades
- Consecutive-tick threshold & alert cooldown configuration
- Unacknowledged filter and bulk-acknowledge workflow

MAINTENANCE SCHEDULER
MTBF-Driven Planning

- Subsystem replacement intervals from ISS MTBF data
- Sensor calibration drift tracking with status flags
- Mission Elapsed Time integrated timeline



Subsystem Replacement Schedule — MTBF-based service cards